ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

2025 SESSIONAL EXAMINATIONS

INTRODUCTION TO DATABASES (CIT2641)

INSTRUCTIONS

(1) Time allowed is **Three (3) hours**.
(2) There are Six (6) questions in this paper. **Answer All Questions in Section A and ANY Two Questions from Section B**
(3) Each question carries **Twenty Five (25)** marks.
(4) Calculators are not allowed.
(5) The total marks for this Exam is **100**

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SECTION A: COMPULSARY QUESTIONS

Question one

a) SQL expresses queries in declarative way. List and explain the six (6) clauses used in expressing SQL queries. **[6 Marks]**

b) Which of these clauses are mandatory? **[2 marks]**

c) Carefully study the following entities and answer the questions that follows:

Department

DeptName	*Address*	*City*
Administration	Bond Street	London
Production	Rue Victor Hugo	Toulouse
Distribution	Pond Road	Brighton
Planning	Bond Street	London
Research	Sunset Street	San José

Employee

FirstName	*Surname*	*Dept*	*Office*	*Salary*	*City*
Mary	Brown	Administration	10	45	London
Charles	White	Production	20	36	Toulouse
Gus	Green	Administration	20	40	Oxford
Jackson	Neri	Distribution	16	45	Dover
Charles	Brown	Planning	14	80	London
Laurence	Chen	Planning	7	73	Worthing
Pauline	Bradshaw	Administration	75	40	Brighton
Alice	Jackson	Production	20	46	Toulouse

i. Find the names of the employees and the cities in which they work **[3 Marks]**

ii. Find the first names and surnames of the employees who work in office number 20 of the Administration department. **[3 Marks]**

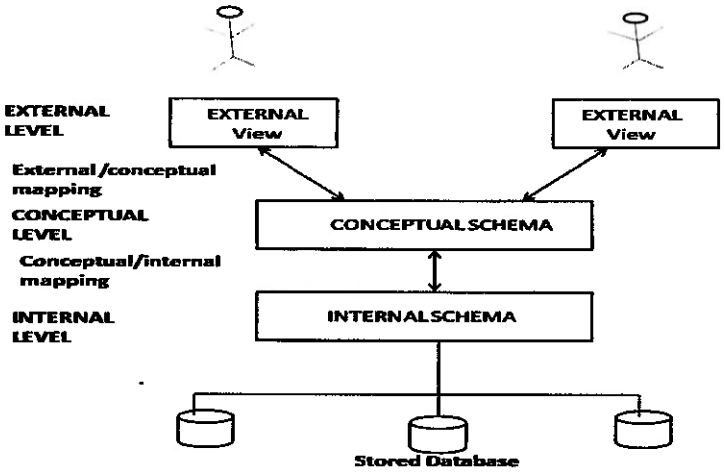
iii. Find the first names and surnames of the employees who work in either the Administration or the Production department. **[3 Marks]**

iv. Find the first names of the employees named Brown who work in the Administration department or the Production department. **[4 Marks]**

v. Find the employees with surnames that have 'r' as the second letter and end in 'n'. **[4 Marks]**

Question Two

a) Consider the following Three Level Architecture and explain how it support different level of data abstraction: **[10 Marks]**



b) Briefly provide a distinction between logical and physical data independence. **[8 Marks]**

c) Compare and contrast Database schema vs. database instance **[7 Marks].**

SECTION B: ANSWER ANY TWO QUESTIONS

Question Three

a) In DBMSs where there is a true three-schema architecture, there are three language types. List and explain these language types. **[6 Marks]**

b) What are the typical functions of Database Management System. **[8 Marks]**

c) There are many tools used to support the database administrator tasks in an enterprise and some of these tools are used during the design phase of a database system. List and explain these tools. **[11 Marks].**

Question Four

a) There are various approaches for data model development. List and explain the most common approaches in use today. **[10 Marks]**

b) Consider a company that is organized into Departments. Each department has a name, number, and an employee who manages the department. We keep track of the start date if the department manager. A department may have several locations. Each department

controls a number of Projects. Each project has a name, number, and is located at a single location. We store each Employees social security number, address, salary, sex, and birth date. Each employee works for one department but may work on several projects. We keep track of the number of hours per week that an employee currently works on each project. We also keep track of the direct supervisor of each employee. Each employee may have a number of Dependents. For each dependent, we keep their name company, set, birth date, and relationship to the employee.

Draw the Entity Relationship Diagram (ERD) to illustrate this Requirements for the Company Database. **[15 Marks]**

Question Five

a) T. Codd (the founder of the relational theory) published a number of rules which attempted to define key features of what a relational DBMS should offer. List and explain these rules. **[12 Marks]**

b) With the use of diagrams explain the differences between Traditional file-based processing and database systems. **[10 Marks]**

c) Briefly explain the various classifications for Database users. **[3 Marks]**

Question Six

Consider the three tables; supplier, shipment and parts. By carefully analyzing the tables, explain the concepts of relational keys. **[25 Marks]**

Supplier

S	S#	SNAME	STATUS	CITY
	S1	Smith	20	London
	S2	Jones	10	Paris
	S3	Blake	30	Paris

Shipment

SP	S#	P#	QTY
	S1	P1	300
	S1	P2	200
	S1	P3	400
	S2	P1	300
	S2	P2	400
	S3	P2	200

Part

SP	P#	PNAME	COLOUR	WEIGHT	CITY
	P1	Nut	Red	12	London
	P2	Bolt	Green	17	Paris
	P3	Screw	Blue	17	Rome
	P4	Screw	Red	14	London